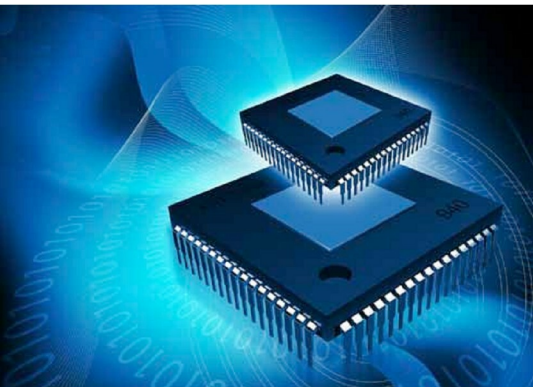


How to Design 2.5/3D Chips: The Future of Electronics

To harness the full capacity of 2.5/3D technology, the engineers need to understand that they will no longer be designing a chip or a circuit individually but a whole system at one shot!



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As 2D chips are getting crowded, 2.5D and 3D chips seem to be the way ahead—to extend Moore's Law. Both involve stacking chips, one above the other, in order to extend the space on chips to the third dimension.

"The main distinction lies in the way chips are stacked together. In 2.5D multi-die packaging, passive silicon interposers are horizontally connected to multiple chips with IC-scale electrical routing between the dies. A full 3D package, on the other hand, uses multiple chips stacked in a vertical manner. In 2.5D, through silicon vas-

(TSVs) within, the silicon interposers are used to route power and signals. In a 3D arrangement, TSVs are used within the active die to create connections from the front to the back of the chips through the silicon substrate," says Giles Peckham, EMEA director marketing, Xilinx.

The 2.5D chips help exceed the expectations of Moore's Law, while also enabling heterogeneity—thanks to the ability to mix and match dies on an interposer and creating 'systems' on that interposer. Many companies and designers are slowly beginning to leverage 2.5D methodology as it can be used without making any significant changes to the exist-

ing design flows and processes.

True 3D integrated circuits (ICs) are expected to offer much bigger benefits. Replacing the traditional wire-bonding process in chip/wafer stacking with emerging TSV interconnection technology would increase inter-die communication bandwidth and reduce form factor besides lowering power consumption of stacked multi-die systems. Hence, true 3D ICs will have the power to overcome the challenges of power, latency, bandwidth, speed and miniaturisation that are prevalent today, and thus will find application in innumerable light-weight, small and low-power devices. Understandably, there is a concerted effort by the industry to move to true 3D, but it is proving to be more difficult than previously thought.

"The distinction between 2.5D and 3D is real and important. There are a number of logistical challenges (test, heat) to implementing true 3D in a cost-effective manner. A major challenge to implement the true 3D design is that its complex functionality would span multiple chips in a stack—rather than having memory on one plane, analogue on another, digital on a third, etc.," points out Guru Ganesan, managing director, ARM, India Operations. "Most top companies are working on these designs at some level, and the concept is very popular, but mainstream design is still very much 2D," Ganesan adds.

Having said that, it is evident that the direction ahead is 2.5D and then 3D, and hence it is wise to be aware of the basics of using such chips in your designs.